Oldham College

Networking Academy

STUDY & ACTIVITY GUIDE

OSI & TCP/IP Trainer

Learn the seven OSI layers and the four-layer TCP/IP model, then drill them with ordering, matching and quiz activities. Best scores are remembered on your device.

The OSI model (7 layers)

#	Layer	PDU	Examples
7	Application	Data	HTTP, DNS, FTP, SMTP
6	Presentation	Data	SSL/TLS, JPEG, ASCII
5	Session	Data	NetBIOS, RPC
4	Transport	Segment	TCP, UDP (ports)
3	Network	Packet	IP, ICMP · Routers
2	Data Link	Frame	Ethernet, ARP, MAC · Switches
1	Physical	Bits	Cables, RJ45, voltages

Mnemonics

- Top → bottom: *All People Seem To Need Data Processing.*
- Bottom → top: *Please Do Not Throw Sausage Pizza Away.*

TCP/IP model (4 layers)

TCP/IP layer	Maps to OSI
Application	5, 6, 7
Transport	4
Internet	3
Network Access	1, 2

Using the trainer

1. **Learn** — read the two models side by side with PDUs, protocols and kit.
2. **Order the layers** — drag (or use ▲▼) to put the seven layers in order, then **Check**.
3. **Match** — drop each protocol/PDU/device onto its correct layer, then **Check**.
4. **Quiz** — ten questions across both models; your best score is saved.

Key idea — encapsulation: going *down* the stack each layer adds its own header (Data → Segment → Packet → Frame → Bits); going *up*, each header is stripped off again.